

Economics journals before and after the financial crisis

A multilayer network analysis

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The project

- Economics is changing, by growing in size and variety, under the impact of research specialization. We try to make sense of this through journals
- Relationships between journals result from the interweaving of conceptual, social and institutional links, which can change over time
 - intellectual communities (based upon contents and cited references)
 - social communities (structured around authors, editors and/or institutions)
- Our goal is to examine patterns of specialization in economics journals, identifying markers of fragmentation within the discipline

A data-driven project

- The unit of analysis are economics journals (nodes)
- As defined in the EconLit list (2019) by the American Economic Association
- The relations between them are defined in reference to various layers of information (edges)
- Available information: **editorial boards** & article **metadata** and **abstracts**
- The first source is **GOELD.db** (Gatekeepers Of Economics Longitudinal Database)
- We integrated it with metadata of the articles

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Jahrgang 1876.

Tübingen, 1876.

Verlag der H. Laupp'schen Buchhandlung.

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AUTUMN '76
VOL. 7. NO. 2.



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Data Collection beyond Editorial Boards

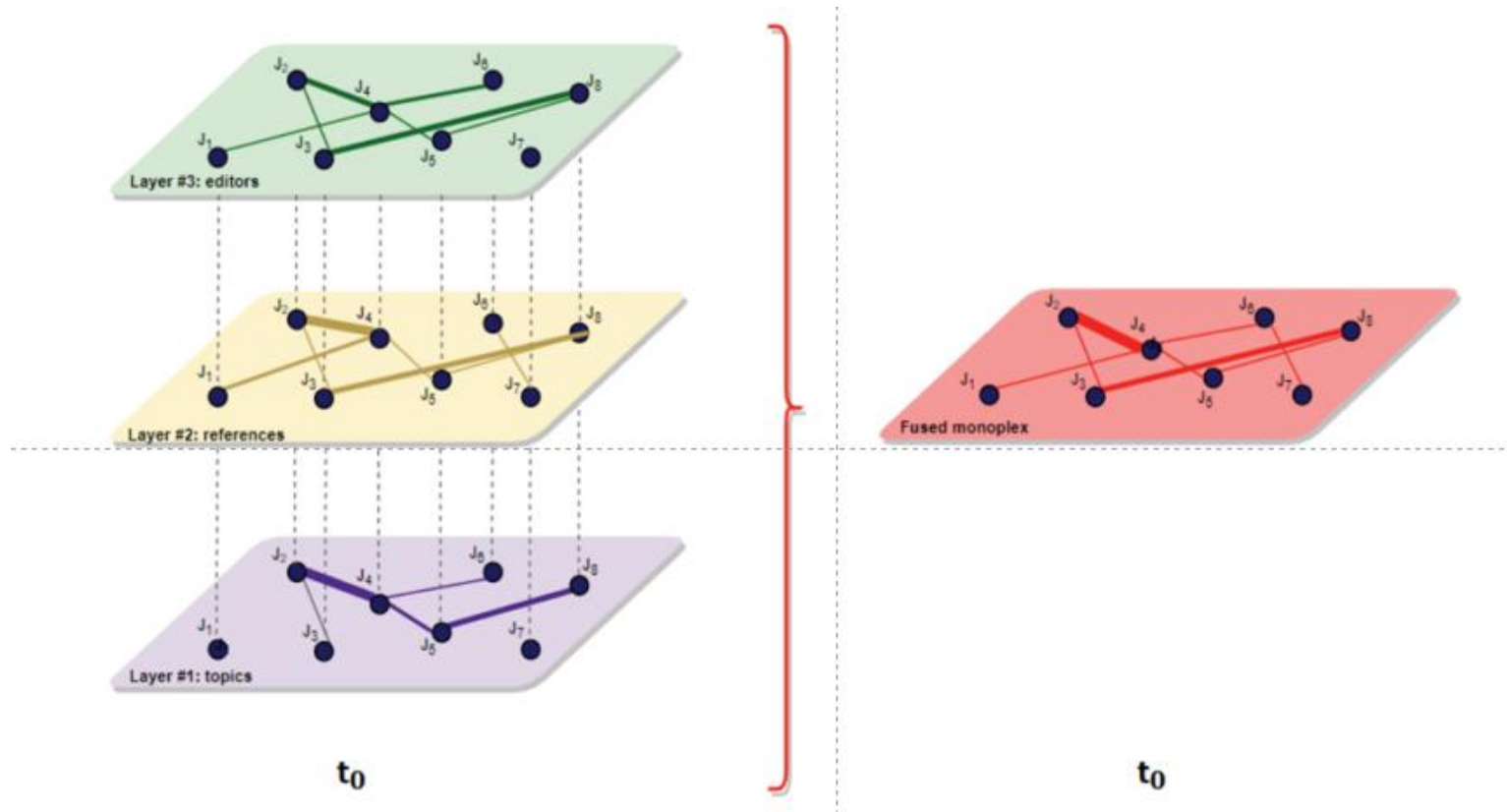
- ISSN and manual matching of journals (1092 unique)
- Source: OpenAlex (wide coverage, free and permissive)
- 3 time windows: 2006-2008, 2012-2014, 2019-2021
- Abstracts: 314 thousand
- References: 3 million
- Authors: 456 thousand



Social relations & intellectual similarity

- Previous works explored these types of relationships (Baccini et al 2020, Baccini, F. et al. 2022)
- The novelty of this project: temporal dimension, text analysis, comprehensiveness
- 4-layer multiplex networks:
 - Editors (journals sharing editorial board members)
 - Authors (journals sharing authorship)
 - References (journals sharing references)
 - Abstract (similarity based on text embedding [SentenceBERT, medoids])
- Similarity network fusion

Multilayer networks and SNF

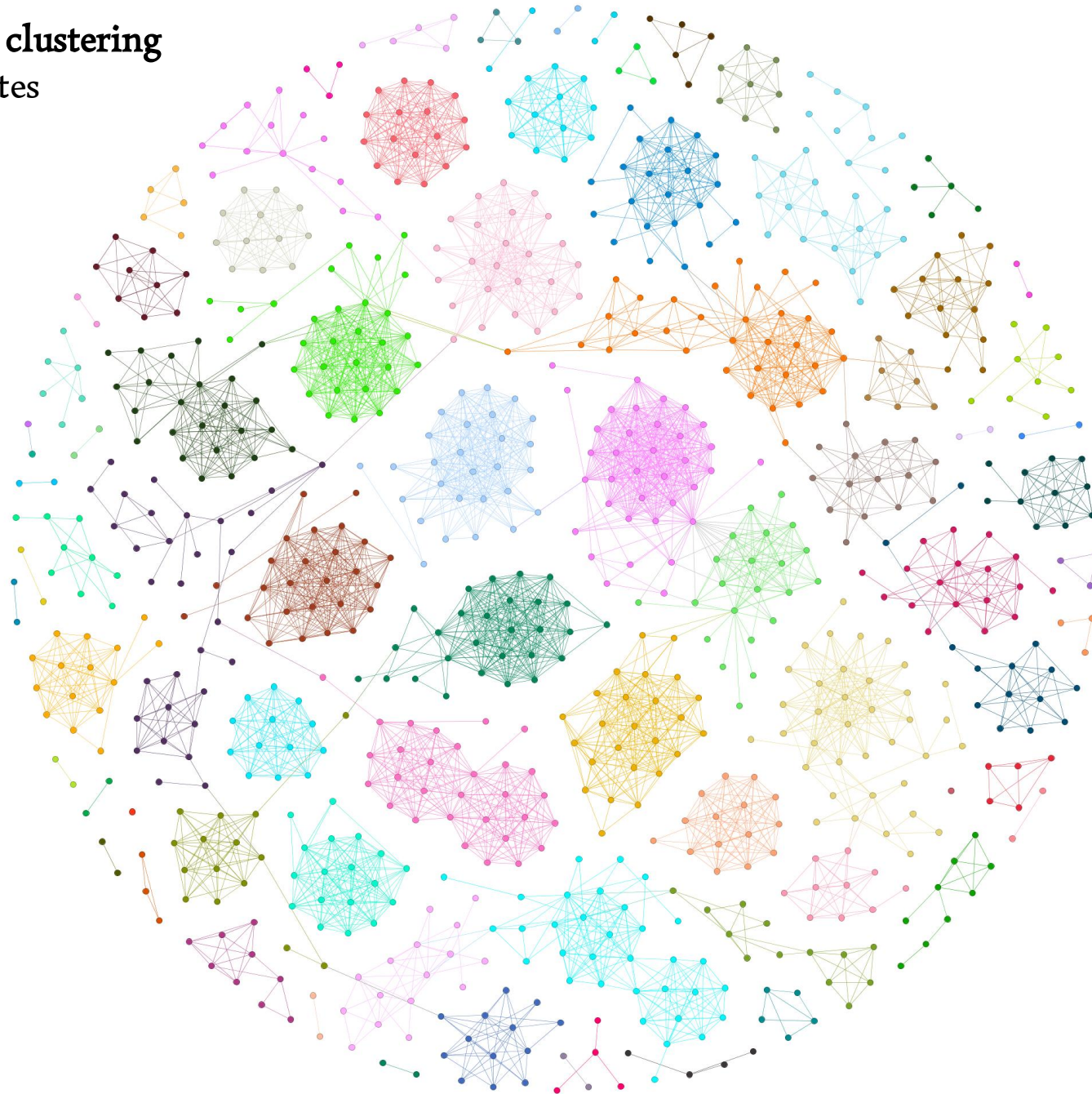


A multiplex network with three layers. Nodes are journals J_1 - J_8 .
On the right the monoplex obtained by fusing three layers

Global structure

- Authors and editors convey nearly identical information, both are quite similar to the references
- Textual content has some relation with the rest but seems to add a different dimension
- Community detection on individual layers results in low modularity, volatile and not very interpretable groups
- Community detection on the **fused network yields better results** and interpretable communities
- Remarkable **stability over time**
- **Ensemble clustering** to identify stable communities

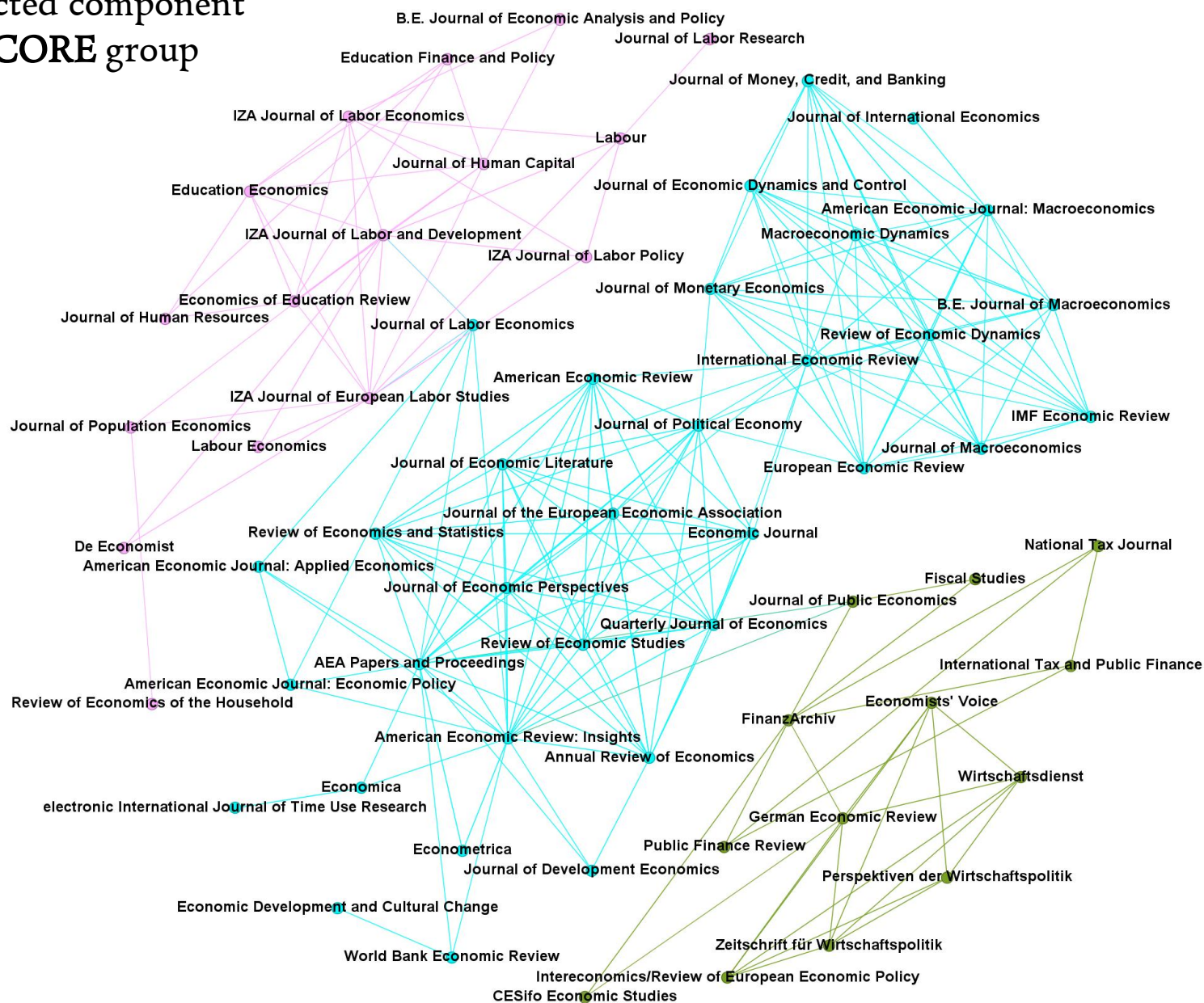
77 groups



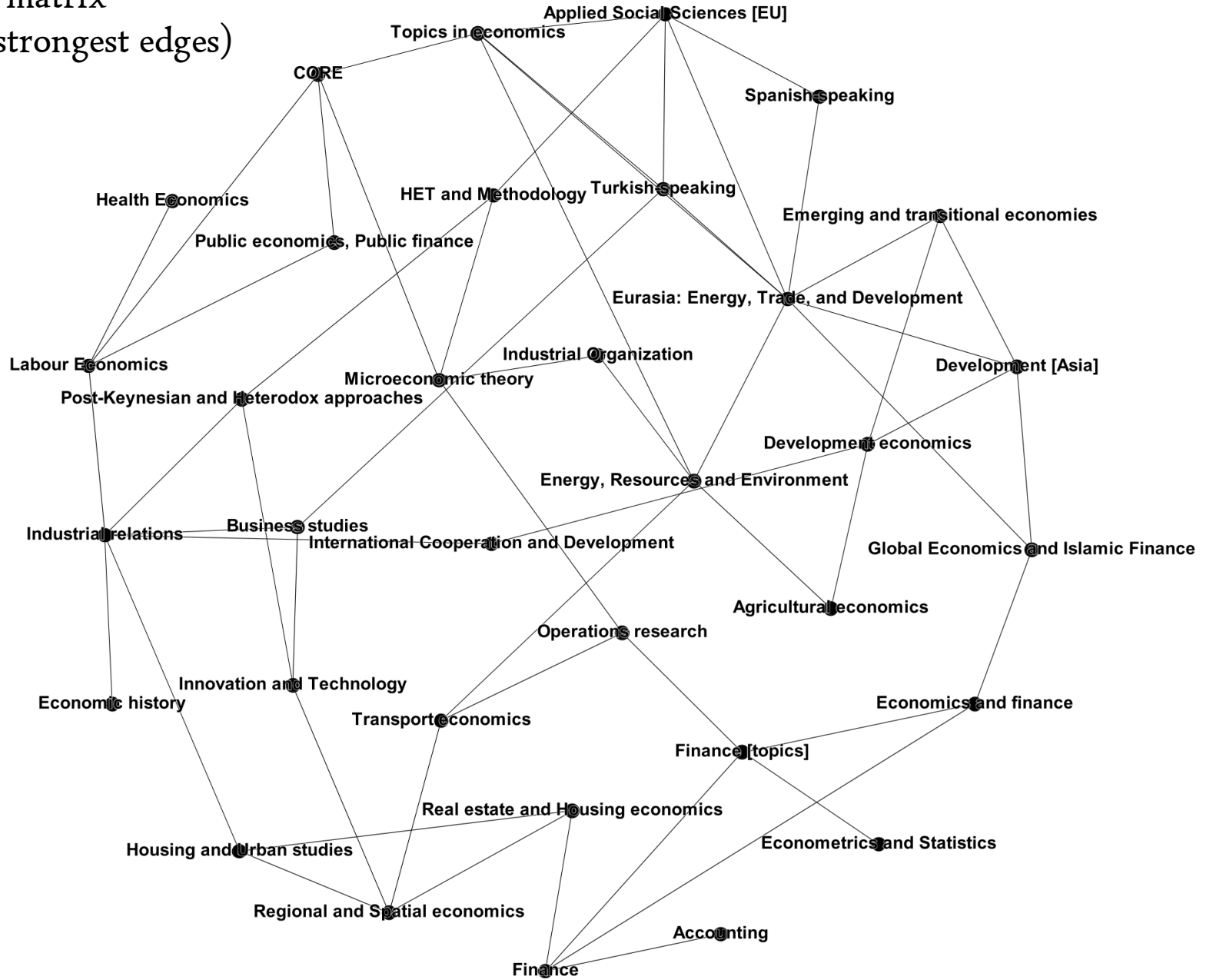
Interpreting journal communities

- Law (1973): theory, technique and subject matter **specialties**.
- Most of our communities clearly fall in the last category
- *Post-keynesian and Heterodox Approaches* is an example of theory-based group (but distinct from HET and Innovation)
- *Econometrics and Statistics* is an example of methods-based
- However, some groups do not fit this narrative:
 - **CORE**: coopting of editors towards the "center"
 - National communities

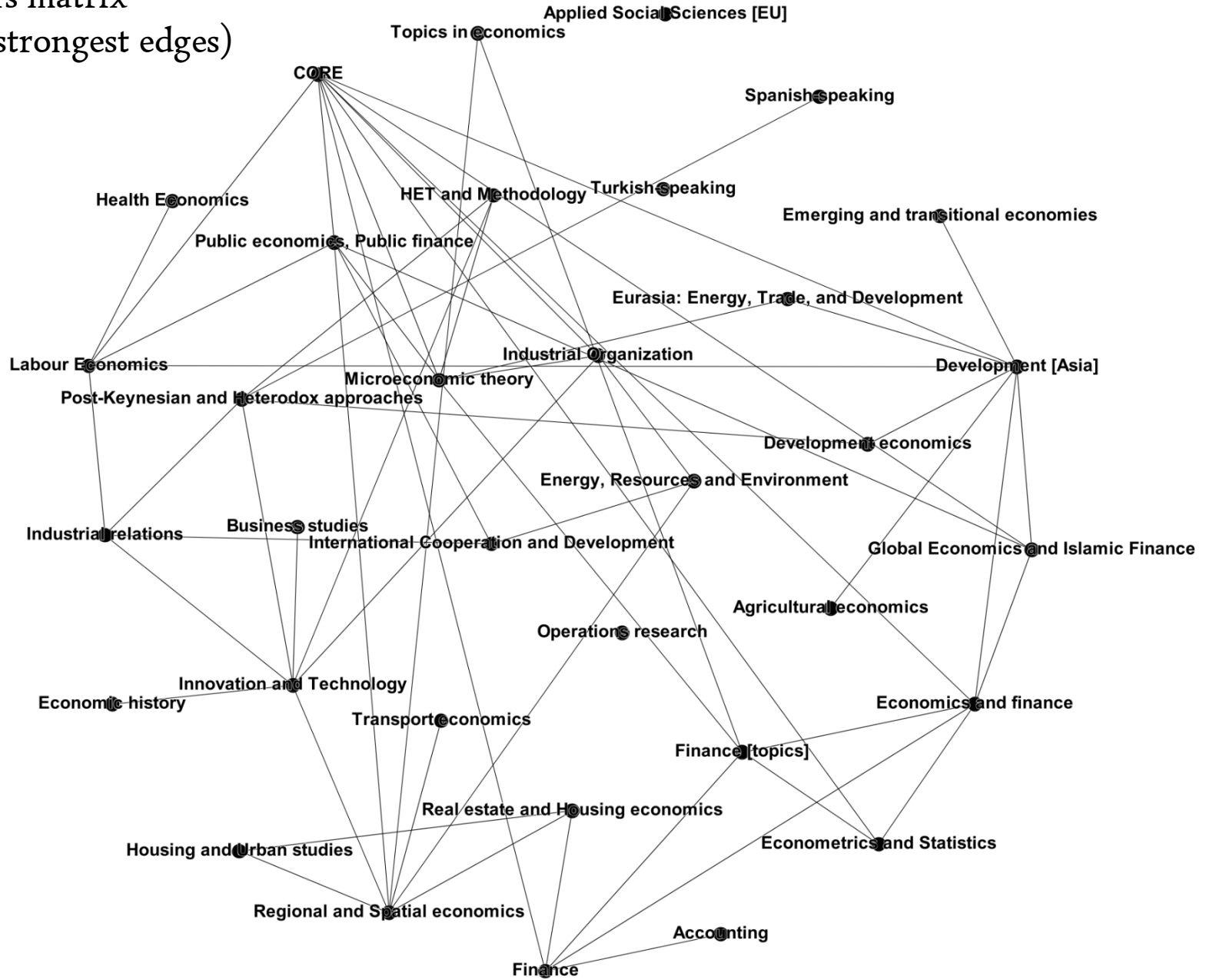
Connected component of the **CORE** group



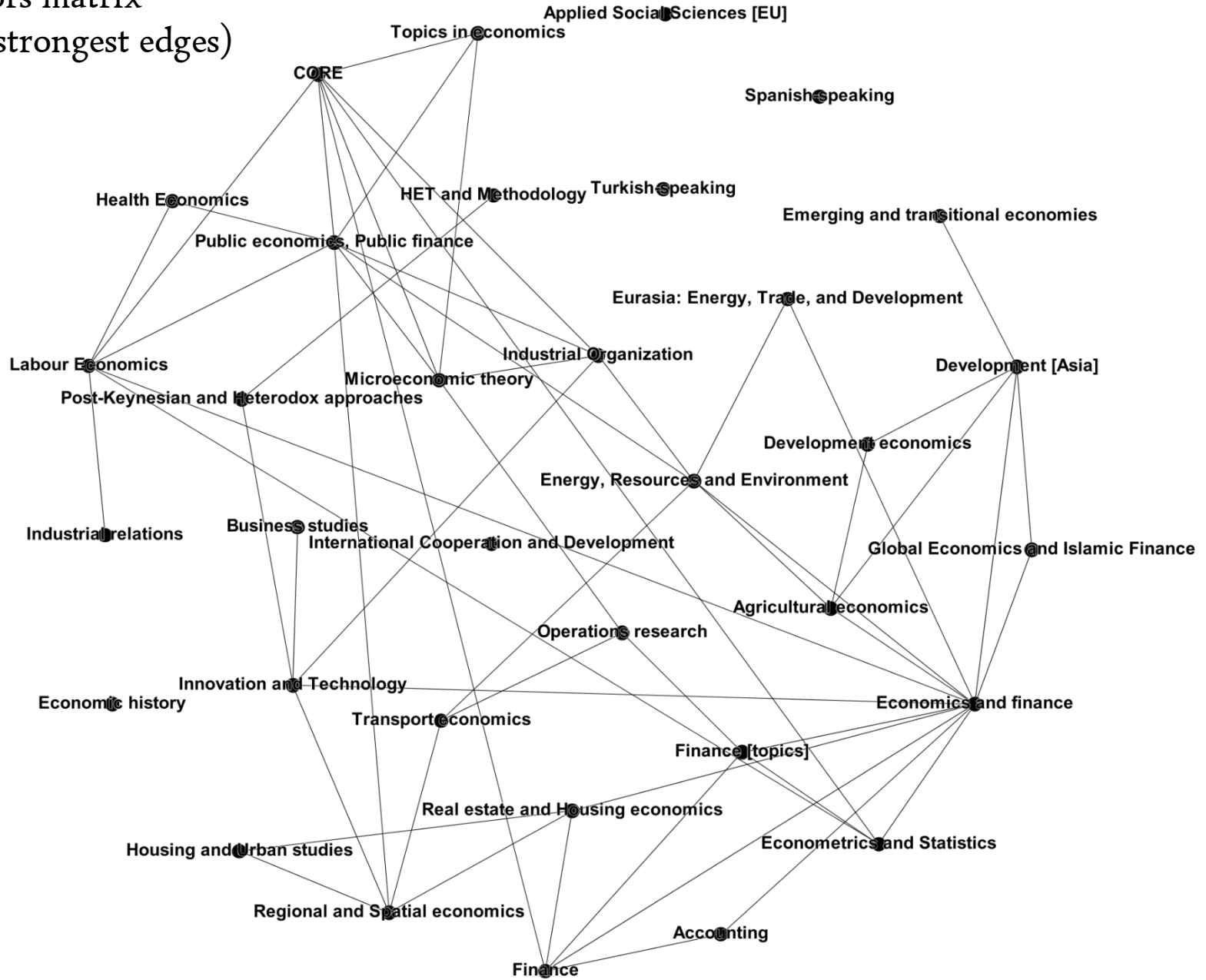
Fused matrix
(10% strongest edges)



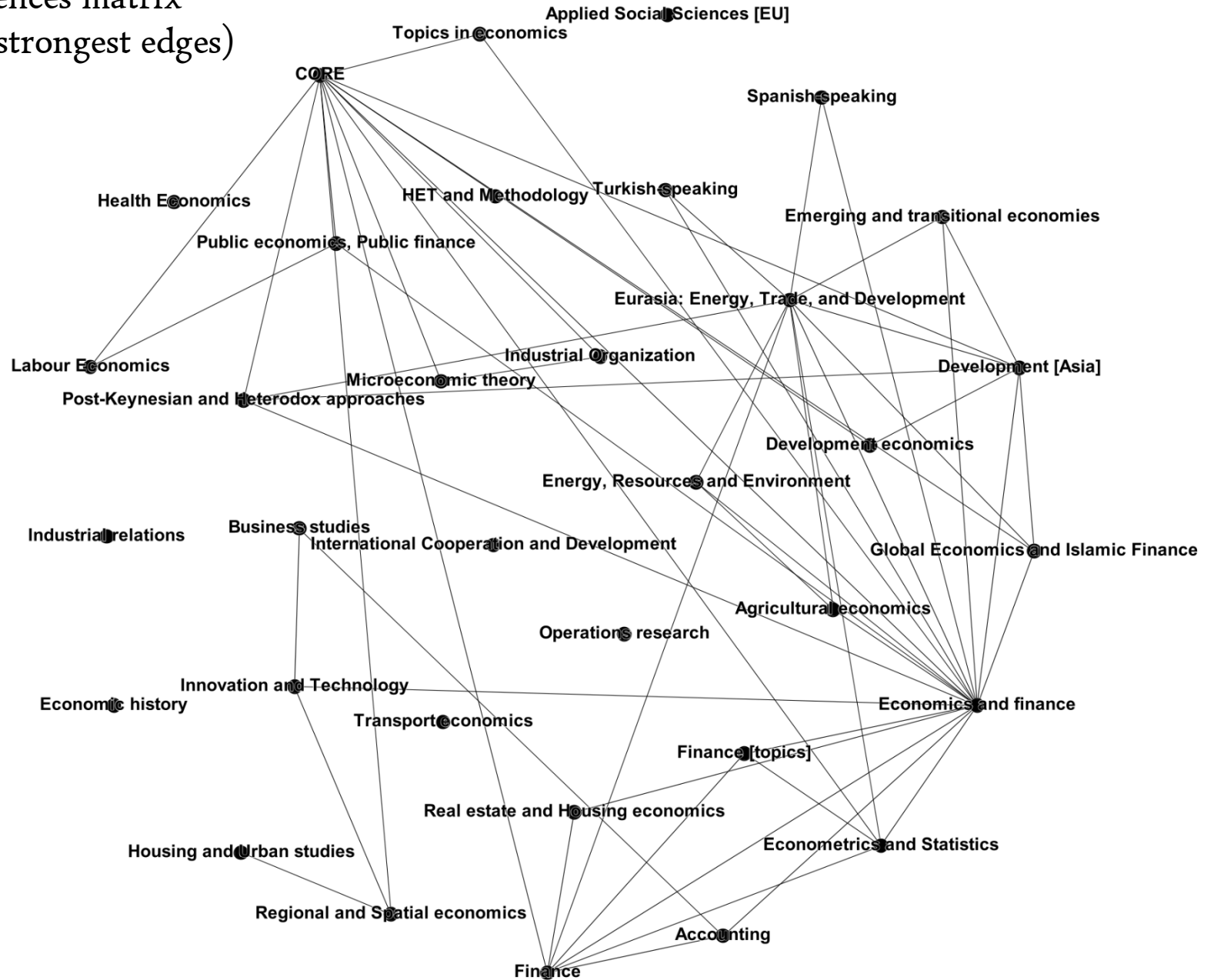
Editors matrix
(10% strongest edges)



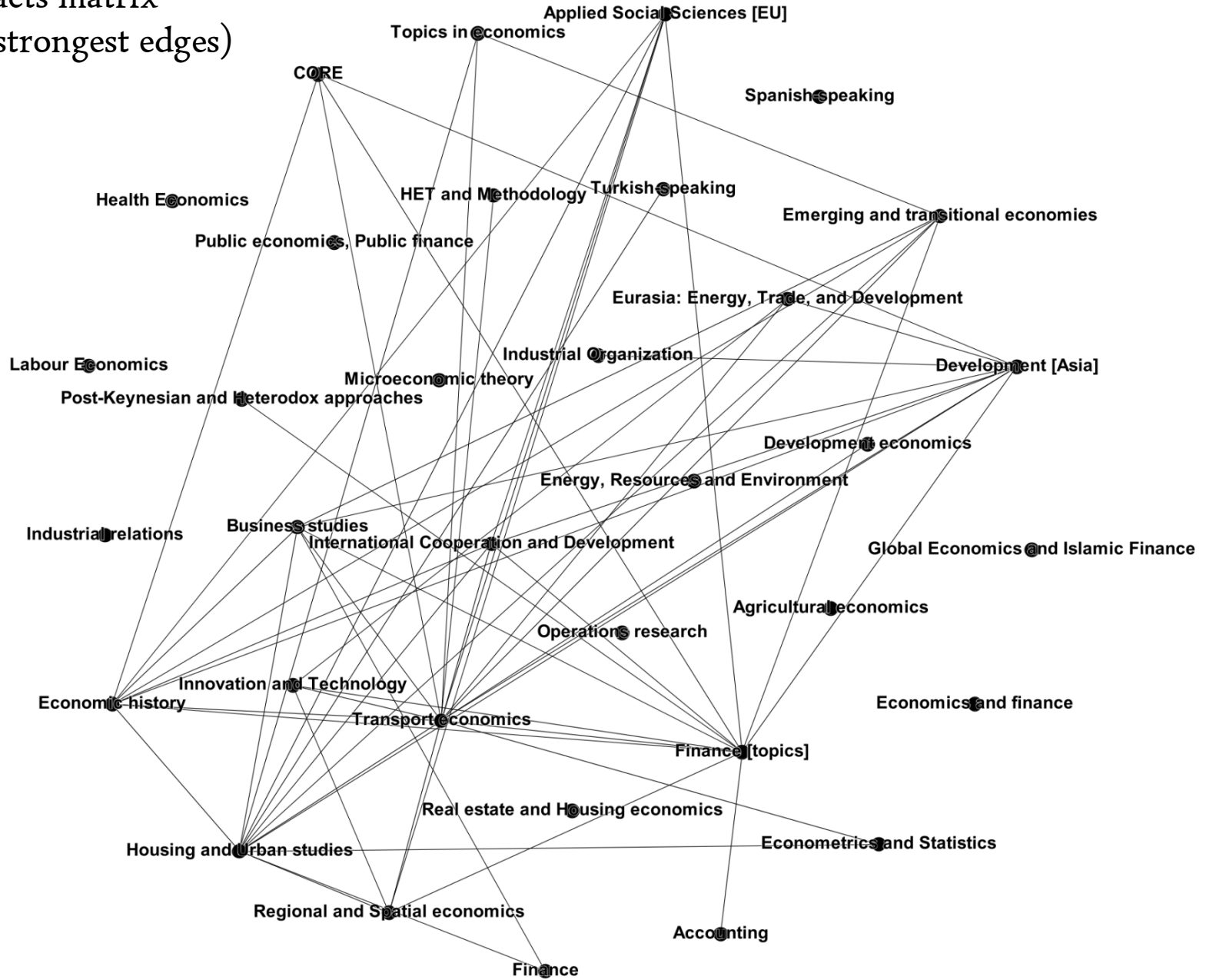
Authors matrix
(10% strongest edges)



References matrix (10% strongest edges)



Abstracts matrix
(10% strongest edges)



Summary

- Not much has changed in the structure of the field before/after the 2008 crisis
- Both the global structure of similarities and the communities are stable
- Combining social and intellectual dimensions provides a richer picture
- Communities can be seen as specialties
- Exceptions to this are CORE and national communities

Thanks!

Comments and suggestion are welcome

Previous works

- Baccini, A., Baccini, F., Barabesi, L., Cioni, M., Petrovich, E., & Pignalosa, D. (2024). Fine-grained classification of journal articles based on multiple layers of information through similarity network fusion: The case of the Cambridge Journal of Economics. *Scientometrics*, 129(1), 373-400. <https://doi.org/10.1007/s11192-023-04884-2>
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